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PlanetX. Manual

Русская версия

Version 0.18.0

PlanetX is a 3D globe inside QGIS. The globe opens in its own window and shows the whole Earth with terrain and atmosphere, from space down to single streets. Layers of the current project, your own placemarks, paths and polygons lie on the globe. Tours fly over the places, and a tour recording gives frames for a video.

Plugin page - www.informpp.ru.

Installation

The plugin runs in QGIS 3.36 and newer, including QGIS 4. It is tested on QGIS 3.36 with Qt 5 and on QGIS 4.0 with Qt 6. It needs a graphics card with OpenGL 3.3 and the Python module PyOpenGL. QGIS builds for Windows include PyOpenGL.

Open **Plugins** → **Manage and Install Plugins**, search for PlanetX and install it. QGIS does not need a restart.

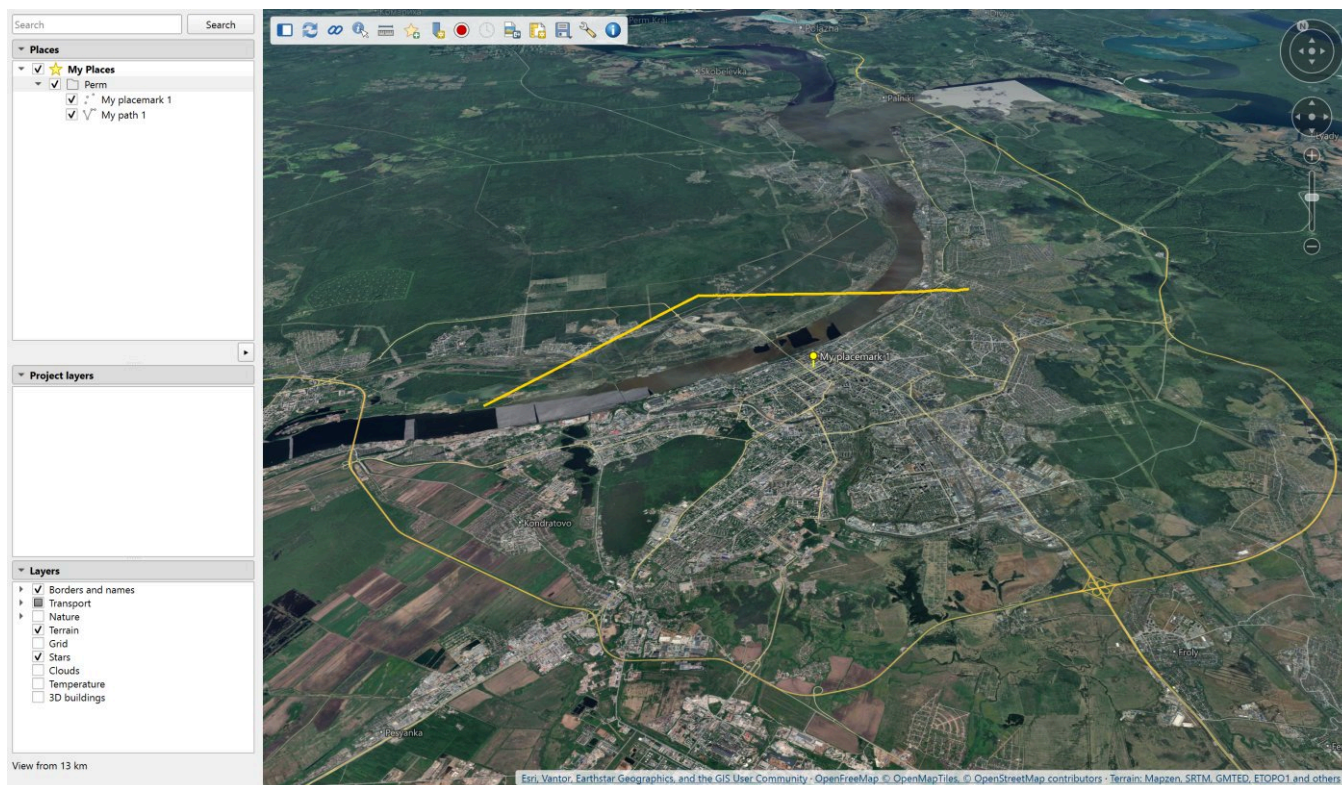
The globe opens with the button on the PlanetX toolbar or from **Web** → **PlanetX**. The same menu holds About PlanetX. If the QGIS build has no PyOpenGL, the plugin says so and does not open the window.

Interface language

The language follows the QGIS settings, not the system. With Russian in QGIS the plugin is in Russian, with any other language it is in English. The label language on the globe is set separately, in the view properties.

The globe window

The panel is on the left, the globe view on the right. The icon bar sits in the top left corner of the view, the data source credits in the bottom right corner. The border between the panel and the view is dragged with the mouse.



Icon bar

Icon	What it does
Sidebar	Hides and shows the left panel, the view takes its place
Refresh	Shows new settings and layers. It turns orange when there is something to show
Synchronization	Links the globe with the QGIS map window
Identify features	Turns on identify by a click on the globe
Ruler	Opens the Ruler window
New placemark	Opens the New placemark window
Save view	Puts a placemark at the look-at point with height, heading and tilt
Record tour	Records the camera movement as a tour into My Places
Time slider	Opens and closes the placemark time slider. It is available when visible placemarks have a time
View snapshot	Saves the view to a PNG or JPEG file
View to layout	Puts the view into a QGIS layout as a picture
Scene	The menu Save Scene... and Open Scene...
Subsurface mode	Drill holes, bed roofs, sections, tunnels and a block cut under the surface, see Subsurface mode
Demo	Ready scenes by body, see Demo
Body	The menu of planets, moons, asteroids and Sky, see Other bodies and Starry sky
View properties	Base map, terrain, labels, link with the map, update, coordinate format
About	Controls, data sources, links

Status line

The status line sits under the left panel. It usually shows the camera height, for example "Overview from 2 000 km". The second line shows the coordinates and elevation of the point under the cursor. The coordinate format is chosen in the view properties.

The line also reports loading:

- "loading 12" - this many tiles, layer pictures and labels still wait for loading.
- "loading has stalled for 7 s" - data waits and nothing has arrived for 5 seconds or longer.
- "places are heavy, 250 thousand vertices" - your own objects exceed 200 thousand vertices, turning may become slow.
- "The base map did not load" and "The vector base did not load" - the source answered with an error.

While loading goes on, a blue icon spins to the right of the icon bar. A short load does not light it, it appears after half a second. When loading stalls, the icon turns orange. Its tooltip tells how much still waits.

Messages of search, scenes, tours and recording stay for 5 seconds.

Navigation

Navigation uses the mouse, the keys and the navigation controls.

Mouse

Action	What happens
Left button drag	The Earth follows the cursor. After release it spins on by inertia
Wheel	Zooms in and out. The point under the cursor stays in place
Left double click	Flies to the point under the cursor and zooms in 2.5 times
Right double click	Zooms out 2.5 times
Right button drag	Up - closer, down - farther, toward the press point
Right click	The menu of the point under the cursor
Middle button or left with Shift	Turn and tilt around the look-at point
Left button with Ctrl	Looks around, the eye stays in place

The camera does not go lower than 50 m above the terrain. The tilt is limited to 85°. While the ruler, drawing or identify is on, a double click stays two plain clicks and places points.

Keys

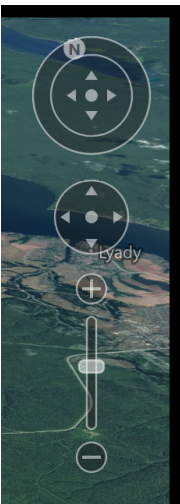
Keys work when the globe view has focus. A click on the globe gives it focus.

Key	What happens
Arrows	Move the view by a twentieth of the window width
Shift and arrows	Turn and tilt by 3°
Ctrl and arrows	Look around by 3°
PageUp, plus	Zoom in by a wheel step
PageDown, minus	Zoom out by a wheel step

Key	What happens
N	North up
U	Look straight down
R	North up and look straight down
Space	Stop inertia and flight

On-screen controls

The navigation controls stand in the top right corner of the view.



Control	What it does
Compass ring	Dragging turns the view. The letter N shows north
Letter N	A click puts north up
Stick inside the ring	While the button is held, the view looks toward the press
Lower stick	While the button is held, the view moves toward the press
Plus and minus	While the button is held, the camera zooms in or out smoothly
Slider	The thumb stands at the camera height, dragging sets a new one

The speed of the sticks grows with the distance from the middle. The controls are always shown. While the cursor is far away, they stand as a faint outline and do not cover the view. When the cursor comes near the corner of the view, they are drawn in full.

Place search

Type a place name into the Search field at the top of the panel, for example Perm, coordinates or a topic, for example "Voyages of Columbus" - this example stands grey in the empty field. Enter or the Search button starts the search. A topic without a place on the map becomes placemarks, see Places from a description.

Format	Example
Decimal degrees	58.0105, 56.2294
Degrees, minutes, seconds	58°00'37.8" N, 56°13'45.8" E or 58 00 37.8 N 56 13 45.8 E
UTM	40V 454464 6430139
MGRS	40V DK 54464 30138

Search also understands the coordinates copied from the status line of the globe together with the elevation.

- Coordinates start a flight at once. The camera lands no higher than 2 km from the point.
- A name goes to the Nominatim service in the label language. The camera flies to the first place found and frames it whole.
- When several places are found, they are listed under the field. A click on a row starts a flight.
- The found place carries a red pin. Clearing the field removes the pin and closes the list.

Suggestions of three kinds appear below the field while typing:

- Placemarks of My Places on the body on the screen with a matching name.
- Earlier queries.
- Stars and constellations when the sky view is open.

A click on a suggestion or Enter on it starts a flight to the placemark or the sky object, an earlier query is searched again. The Down key moves into the list of suggestions, in an empty field it shows the earlier queries. Escape closes the list. Place names from the network are not suggested while typing. The Nominatim service takes a query only on Enter.

Enter in the field handles the query in this order:

1. Coordinates start a flight.
2. The exact name of an own placemark starts a flight to it with no network request. In the sky view the name of a star or a constellation works the same way.
3. A request in words goes to the assistant, the Assistant section describes it.
4. Anything else is searched as a place name.

Queries entered with Enter are kept in the QGIS profile, 30 at most. The Clear search history button of the Assistant group in the View properties window erases them.

Over a long distance the camera rises and lands smoothly. The mouse interrupts a flight.

Left panel

The panel has three sections - Places, Project layers and Layers. A click on a header collapses the section, and the open sections take its place. The globe remembers which sections are collapsed.

Layers section

Its check boxes take effect at once, without the Refresh button.

Group	Rows
Base map	Sources of imagery and maps, one is checked, the last row is Add tile source...
Terrain	Mapzen Terrain Tiles elevations and hill shading
Borders and names	Borders, Places, Water names
Transport	Roads, Road numbers, Railways, Airports
Nature	Rivers, Lakes and reservoirs, Peaks, Reserves and national parks
Grid	Parallels and meridians with labels, the equator, tropics and polar circles in yellow
Stars	Stars brighter than magnitude 6 and the Milky Way at their places in the sky

Group	Rows
Clouds	Clouds from NASA GIBS VIIRS imagery of the last complete day
Temperature	Surface temperature of land by day over 8 days (MODIS) and of the sea over a day (GHRSSST MUR) with a scale in degrees
3D buildings	OpenStreetMap buildings as blocks from OpenFreeMap tiles, off by default
Sun	Light of the terrain, buildings and air by the position of the sun, the night side of the Earth is dark with city lights on it, off by default
Earthquakes	Earthquake foci of magnitude 4.5 and above over 30 days from the USGS feed, off by default
Earth cutaway	A sector removed from the Earth with the shells on its faces, off by default
Slope	Surface slope from the heights in classes from flat to steeper than 35°, off by default
Aspect	The direction a slope faces, in the colours of the compass directions, switched on instead of the slope

The vector base comes from OpenFreeMap tiles. On the first opening borders, places, terrain and stars are on. The tooltip of each row tells how the feature is drawn and from what height it is visible.

The grid step follows the camera height, from 30° from space to seconds near the ground. The stars stand where they stand over the Earth at this minute and fade when the camera goes below 150 km. The Milky Way image, about 10 MB, is downloaded when the stars are first shown and comes from the QGIS cache afterwards. Clouds lie as a veil over the imagery. Snow and ice get into the clouds too, they have the same colour.

Temperature colours land and sea with their own scales, the scales in degrees Celsius are in the bottom left corner of the view. Land shows the temperature of the surface itself by day, not of the air, over 8 days. Land may have gaps under clouds. Sea shows the water temperature near the surface over a day.

The Sun row lights the terrain and buildings from the side of the sun. The night side of the Earth is dark, the air over it does not glow. City lights shine on it - the NASA Black Marble image of 2016, pixel about 600 m. In the twilight the lights fade together with the night, under clouds they are weaker. The sun time is the end of the interval of the open time slider, so the light of the hour of a placemark shows. Without the slider the sun follows the computer clock. Without the Sun row the light falls from the north-west at 45°, as on a relief map. The position of the sun is computed to about 0.01°, there are no shadows.

Labels stay level at any turn and tilt and do not overlap. A place behind a mountain or beyond the horizon has no label.

The Slope and Aspect rows colour the surface by the terrain heights. Slope is the angle of the surface to the horizontal, in classes 0-2°, 2-5°, 5-10°, 10-15°, 15-25°, 25-35° and steeper than 35°, from green to purple. Aspect is the compass direction a slope faces downhill, each of the eight directions has its colour, ground flatter than 0.5° is grey. The scale is in the bottom left corner of the view. The slope is computed from the true heights, without the vertical exaggeration of the view. The layer exists on the Earth, Mars and the Moon, other bodies have no heights. The accuracy is limited by the height pixel. It is about 5 m at the equator on the most detailed level on the Earth, 2.6 km on Mars, 1.3 km on the Moon.

Sea and ocean depths are part of the terrain. The floor lies at its depths, and semi-transparent water lies above it at sea level. Water shallower than 200 m is more transparent, the floor shows near the shore. Below sea level the status bar shows the depth under the cursor. The depths come from the same Terrarium tiles as the land heights, in the ocean these are ETOPO1 data. Sea names stay at sea level. The elevation profile, the ruler and the slope use the floor, the viewshed and insolation use the water surface. A profile below sea level shows the water and a dashed sea level line. Dry depressions below zero, for example the Dead Sea, are not

told apart from the sea by the heights and are filled on the profile too. The Sea and ocean depths box in the Terrain group of the view properties switches them off, then heights below sea level count as zero and the sea is flat. Depths exist only on the Earth.

The Paleogeography row shows the Earth in the past, up to 540 million years ago, after the PALEOMAP PaleoDEM maps (Scotese and Wright, 2018). The map of an age holds land heights and sea depths on a 0.1° grid, about 11 km, coloured by height with hill shading. A slider in the bottom left corner of the view sets the age in millions of years in steps of 5 million years, the geological period is named next to it. The ► button shows the ages one after another towards the present. Today's imagery, borders, labels and relief are removed for that time, the relief setting itself does not change. The maps lie as tiles in the planetx-terrain repository, the first showing of an age needs the internet. A new age appears at once in a coarse form and sharpens as it loads. When the window opens, the row is off.

The Planet of Fire, Planet of Water, Gases and Land and Life groups at the bottom of the Layers section hold NASA themes. A theme is a raster by days, months or years from the NASA GIBS service. The groups hold:

- Planet of Fire - fire spots, smoke, aerosol optical depth, carbon monoxide and its emission.
- Planet of Water - precipitation, soil moisture, snow, sea ice, water vapour, chlorophyll, sea salinity and floods.
- Gases - nitrogen dioxide, sulphur dioxide, methane, carbon dioxide and ozone.
- Land and Life - vegetation, dust, night lights and land cover.

Themes are radio buttons, one theme of all groups is on. Checking another theme clears the previous one. The group check box is checked while its theme is on. Clearing it turns the theme off, checking it turns the previous theme of the group on. The theme lies as semi-transparent colouring over the imagery, where there is no data the imagery shows. A legend with the name, units and date stands in the bottom left corner of the view. The day of the theme is set by the right handle of the time slider, see Placemark time and the time slider. A chosen theme opens the slider on the last day of its series. The handle gives the nearest day of the series that is not later, missing days of the series are skipped. A new theme takes the same moment while the slider is open. Some series lag behind today by months, the tooltip of the theme says so. Themes exist only on the Earth.

The first row of the Planet of Fire is Fires, a radio button in the same choice as the themes: either the fires or one theme is on. The row shows fire spots over the last 24 hours from VIIRS images of the NOAA-20 satellite from the NASA FIRMS feed, about 78 thousand spots. A dot stands at the fire, its colour and size show the radiative power from 1 to 1000 MW, the legend stands in the bottom left corner of the view. The feed, about 6 MB, is loaded each time the row is switched on. The spots stand on the time slider by the image time. A click on a spot in the Identify mode shows its power, image time and confidence.

The Plate boundaries row shows the lithospheric plate boundaries after the PB2002 model (Bird, 2003). Red lines are plates moving apart at oceanic ridges and continental rifts, green - plates sliding along transform faults, blue - plates converging in subduction and collision zones. Plate names show as labels. A click near a boundary in the Identify mode shows the boundary type, the plate pair and the speed of their relative motion in mm/yr. In the plate pair a slash in Bird's notation shows which plate goes under which: "PA\OK" is the Pacific plate under the Okhotsk plate.

The Earthquakes row shows earthquakes of magnitude 4.5 and above over the last 30 days from the feed of the U.S. Geological Survey (USGS). The feed is loaded each time the row is switched on. A circle marks the focus at its depth, and a thin line leads from it to the epicentre on the surface. The colour of the circle shows the focus depth, from red near the surface to purple at 700 km, the scale is in the bottom left corner of the view. The circle grows with the magnitude. Foci show through the surface, foci on the far side of the Earth are

hidden. The depth is stretched by the vertical exaggeration of the terrain, like the heights. Each event has a time. With the row on, the time slider becomes available, and it shows the events of the chosen interval. The row exists only on the Earth.

The Earth cutaway row removes a sector of the Earth under the look-at point. The sector is a quarter of a hemisphere 90° wide in longitude, from the equator to the pole of the hemisphere of the look-at point. Its three faces are coloured by shells after the radii of the PREM model. These are the crust to 24.4 km, the upper mantle to 400 km, the transition zone to 670 km, the lower mantle to 2891 km, the outer core to 5149.5 km and the inner core. The shell scale is in the bottom left corner of the view. The edge of the faces lies on the ellipsoid, the depths are not stretched by the vertical exaggeration. The crust on the faces comes from the CRUST1.0 model in 1° cells. These are water, ice, upper, middle and lower sediments, upper, middle and lower crust, below the Moho the mantle begins. The crust is 6-7 km thick under the ocean and about 70 km under Tibet. Where a face crosses a subduction zone, the subducting slab from the Slab2 model of the U.S. Geological Survey shows on it as a band from the upper surface of the slab down by its thickness. Slabs exist in 27 zones. The file of a zone is requested when a face touches it, the loading icon shows this. The model is downloaded from the site of its authors when the row is first switched on, until then the crust is the 24.4 km PREM layer. The crust layers can be told apart up close, where the depths on the faces are true. As the camera moves away, depths to 400 km are stretched so that the crust stays visible. At the surface the stretch reaches 8 times, towards 400 km it fades out, below the scale is true. The shell scale shows the factor, earthquake foci are stretched the same way. Labels of places in the removed sector are not shown. The sector is placed anew each time the row is switched on. The corners of the sector are marked with points showing their latitude and longitude, and they can be dragged with the mouse. A corner changes its own longitude and its own latitude. If the latitude is neither 0° nor a pole, the lower or upper face becomes a cone towards the centre of the Earth. The row exists only on the Earth.

The Section down... item in the menu of a path in My Places opens the Section window. It shows a section of the Earth down along the path. The distance from the start of the path runs across, the depth runs down. The section shows the CRUST1.0 crust layers, the PREM shells, the Slab2 slabs and earthquake foci from a band along the path. The colours are those of the Earth cutaway faces, the layer scale is to the right of the chart. The Depth list sets the depth of the section, from 100 km to the centre of the Earth. The Foci band field sets the width of the band. A focus is moved onto the section at the nearest point of the path. The cursor over the chart marks the point on the globe and shows the Moho and slab depths there. The crust, the slabs and the earthquake feed are loaded when the window opens, if they are not there yet, the line under the chart tells what is still loading. While the window is open, a wall of the same section hangs along the path on the globe. It shows through the surface, its depths are true. When the window closes, the wall disappears.

Base map and a tile source of your own

The Base map group lists Esri World Imagery, OpenStreetMap and the XYZ Tiles connections of the QGIS browser. Checking a row changes the base map on the globe at once. On other bodies the group shows the imagery of the body, the choice is not available.

The row Add tile source... opens the new source window. The Address field takes an address in any of these forms:

- a template with {z}, {x}, {y}, for example <https://{s}.tile.opentopomap.org/{z}/{x}/{y}.png>
- the address of one tile copied from a browser, for example
https://server.arcgisonline.com/ArcGIS/rest/services/World_Topo_Map/MapServer/tile/5/10/17
- an address with the level, column and row in the query

The window turns the address into a template and suggests a name from the server name. Below it builds a mosaic of the whole world from four level 1 tiles. If north is at the bottom of the mosaic, the server rows go from the bottom up, the box Rows from the bottom up (TMS) fixes this. The window finds the most detailed level itself by asking for tiles at the look point of the globe. No-data placeholders do not count. The source credit is in the corner of the view and on snapshots, for known servers it is filled in by itself.

The source is saved as a QGIS XYZ Tiles connection and becomes the base map at once. It is also visible in the QGIS browser, where it can be edited or removed. The owner of the tiles sets their terms of use.

3D buildings

The 3D buildings row of the Layers section puts OpenStreetMap buildings on the globe as blocks. Outlines and heights come from the same OpenFreeMap vector tiles as the vector base. The row is off by default.



Buildings show when the camera is closer than 6 km to the ground. The globe keeps up to 32 building tiles around the camera, the nearest first. Tiles behind the camera are not loaded.

The height of a building is its height in OpenStreetMap, otherwise the number of floors times 3.66 m. A building without either gets a height of 5 m. In the centre of Perm two thirds of the buildings have a height of their own, in the centre of Moscow almost all. In the outskirts many houses have the same height. The colour of a building comes from OpenStreetMap, otherwise the building is light grey. Roofs are flat.

The bottom of a building stands at the lowest point of the terrain under its outline. A view snapshot waits until all buildings of the frame are loaded. A scene remembers whether buildings are on.

View properties

Base map

Esri World Imagery - example

Terrain

Vertical exaggeration 1.0

Labels

Language As in QGIS

QGIS map

Synchronization Both ways

☐ Layers as on the QGIS map

☐ New layers straight to the globe

Update

☐ Update automatically

Coordinates

Format Decimal degrees

Заккрыть

The View properties icon on the icon bar opens the View properties window. The window does not block work with the globe.

Group	Field	What it sets
Base map	List	Source of imagery or map
Terrain	Vertical exaggeration	Height multiplier from 0.5 to 10
Terrain	Sea and ocean depths	The floor at its depths with water above it, on by default
Labels	Language	Language of labels and search
QGIS map	Synchronization	Direction of the link with the map
QGIS map	Layers as on the QGIS map	The globe shows the layers switched on in the QGIS legend
QGIS map	New layers straight to the globe	A new project layer is checked on the globe at once
Update	Update automatically	The globe refreshes after every change without the Refresh button, on by default
Coordinates	Format	Decimal degrees, degrees-minutes-seconds, UTM or MGRS in the status line and the Features window. North of 84° and south of 80° UTM and MGRS are replaced with decimal degrees

The default base map is Esri World Imagery, listed as an example. Esri sets its terms of use. OpenStreetMap comes next, then the XYZ Tiles connections from the QGIS browser. Terrain connections are not listed. A new connection appears in the list the next time the window opens.

The Data sources... button next to the base map list opens a window with all globe sources. Its table lists by groups the base maps, terrain, vector base, NASA data and Earth science data. Each source shows what it gives and an open link to its terms of use. The Check button requests one tile or file from each source bypassing the cache and writes the answer time and size or an error. Large files, the star sky image, are not requested.

The same window sets own terrain - a height tile template with {z}, {x}, {y} in Terrarium or Mapbox Terrain-RGB encoding - and an own vector base - a TileJSON address in the OpenMapTiles schema. Each has a credit for the view corner. Apply takes the data from the new address, Reset returns the default source. The Add base map..., Edit... and Remove buttons work with the QGIS XYZ Tiles connections, a removed connection disappears from the QGIS browser too.

The label language is chosen from a list. As in QGIS takes the QGIS interface language, Local names keep the names in the language of the country. Fifteen languages follow. When there is no name in the chosen language, Latin-script languages get the Latin name, the others the local one.

A change of base map, terrain exaggeration or project layers shows after the Refresh button on the icon bar. The status line reminds about it. The label language changes at once.

Project layers section

The section lists the layers of the current project in the order of the QGIS map, each with its type - raster, points, lines, polygons or table. The check box of a layer shows it on the globe and does not change its visibility on the map. In a new project the boxes are clear. The checks are stored in the project.

Layers lie on the globe as a picture drawn by QGIS, with the styles of the map. The picture lies on the terrain, it cannot be raised or extruded.

Labels of vector layers come separately, like city names. They stay level at any turn and tilt, do not overlap and hide behind mountains and the horizon. The text, colour and size come from the layer labels in QGIS, rule-based labels too. A point label stands to the right of the point, a line label at its middle, a polygon label inside it. Features around the view point are labelled, at most 500 per layer. The scale visibility of QGIS labels is not used on the globe.

A double click on a layer flies to its extent, looking straight down. The right-click menu:

- Fly to - a flight to the extent of the layer.
- Transparency slider - the transparency of the QGIS layer itself, it changes on the map too.
- Track... - for point layers, see [Tracks](#).
- Globe terrain - for rasters, see [Own terrain](#).
- Layer Properties... - the standard QGIS layer properties window.

Own terrain

A height raster of the QGIS project becomes the globe terrain within its extent. An example is a quarry with a mine survey. The Globe terrain item of the raster menu in the Project layers section turns the inset on and off. The raster can be in any coordinate system, it needs a file that GDAL reads. The checkbox that shows the layer on the globe is not needed for the inset.

Where the raster has data, its heights replace the common terrain. Along a band at the edge of the data the heights change smoothly to the common terrain, the band is 5 % of the smaller side of the extent. Within the extent the terrain is more detailed than the common one, down to the raster pixel, but not finer than the pixel of level 17 (about 0.6 m at latitude 60°). The raster heights are taken as they are. If the survey has its own height system, the band at the edge smooths the difference with the common terrain.

The inset is seen by the surface mesh, the height under the cursor, the ruler, the elevation profile, the viewshed, the insolation, the slope and the aspect. With several insets the one higher in the project lies on top. The choice is kept in the project and in the scene.

On the first switch-on the raster is converted into a file of the QGIS profile. For a survey of 2400×2400 pixels this takes about 2 s.

Places section

The section holds the My Places folder, described in the chapter [My Places](#).

Link with the QGIS map

Synchronization

The synchronization icon on the icon bar links the globe with the QGIS map window. The direction is chosen in the view properties:

- Both ways - the default.
- Map leads the globe - the globe moves to the map extent and keeps its tilt and heading.
- Globe leads the map - the map centers on the look-at point of the globe once the globe stops.

Any map coordinate system works. The map width on the ground equals the width of the globe view.

Selection

Features selected on the QGIS map are highlighted on the globe.

Identify features

The Identify features icon turns identify on, the cursor becomes a cross. A click on the globe puts a pin with the coordinates and opens the Features window.

- The top of the window shows the coordinates and elevation of the point.
- Below are the project layers checked on the globe, their features under the point and the feature fields. A raster shows its values at the point.
- The Select on map button selects the found features in the QGIS layers. The selection shows on the map, in the attribute table and on the globe.
- The My Places group lists the visible placemarks, paths and polygons under the point. Each shows its type, coordinates or length, measurement and description.
- The Earthquakes group lists the foci near the click point on the screen. Each shows its magnitude, depth, UTC time, place and the USGS event page.
- The Site group shows the elevation or sea depth, the Moho depth, the crust and sediment thickness after CRUST1.0 and the Slab2 slab under the point. The crust model and the slab zone file load on the first click, the window updates when they arrive.

My Places

My Places is the folder at the top of the Places section, marked with a star. It holds placemarks, paths, polygons, saved views and measurements. The places are kept in one file `PlanetX/myplaces.gpkg` in the QGIS profile folder and are visible in any project.

List rows

The row icon shows the kind of object - placemark, path, polygon, saved view or folder. A check box shows and hides the object on the globe, a folder check box does it for the whole folder. A double click on a place flies there.

Places and folders are rearranged by dragging, including from folder to folder. The tour follows the list order.

Several rows are selected with Ctrl and Shift. The selection is dragged together, and the Del key deletes it after a question.

Menus

Where	Items
My Places	Add, Play tour, Sort A-Z, Open KML or KMZ..., Save as KML..., Copy, Paste, Add the places layers to the project, Clear My Places...
Folder	Fly to, Add, Cut, Copy, Paste, Delete, Open KML or KMZ..., Save as KML..., Snapshot folder view, Sort A-Z, Play tour, Properties...
Place	Fly to, Tour along the path, Snapshot view, Properties..., New Folder, Cut, Copy, Paste, Delete
Several rows	Copy, Show selected, Hide selected, Delete selected
Empty space	New Folder, Open KML or KMZ..., Paste

Only paths have Tour along the path. The Add submenu creates a folder, placemark, path, polygon or recorded tour in the folder. The placemark, path and polygon are drawn in the New place window, the tour is recorded with the record button. Cut copies the row to the clipboard and removes it from the list, Paste puts it in a new place. Sort A-Z orders places and folders by name.

Folder properties

The Folder properties window opens from the folder menu:

Field	What it sets
Name	The folder name in the list
Allow the folder to be expanded	Without the box the folder does not expand in the list, the folder box shows and hides all its contents
Show contents as option buttons	One row of the folder shows on the globe, checking one clears the others
Description	Text about the folder, the tooltip of its row
View	Look point, range, heading and tilt, buttons Snapshot current view and Reset

A double click on a folder and Fly to in its menu fly to the folder view. Without a view of its own the flight frames all places of the folder. The view is set by hand with numbers on the View tab or by a snapshot. Snapshot folder view sets the folder view to the view of the globe. The description, the view and the way contents show go to KML and back.

A new folder appears at once with the name New Folder, the name is changed in the Properties... window. On a folder it is created inside, on a place right below it. A folder is deleted with its contents after a question. Add the places layers to the project adds the three layers of the places file to the project - points, lines and polygons. Their attributes and undo are the standard ones.

New placemark

The New placemark icon opens a window with the tabs Placemark, Path and Polygon.

- A placemark is put with a click on the globe, a new click moves it.
- Path points and polygon vertices are put with clicks on the globe.
- The Name field is filled at once with a numbered name - My placemark 1, My path 1, My polygon 1. Each kind counts on its own. The name can be changed in the same field.
- Color sets the color of the line and outline, the polygon fill has the same color, half transparent. Width is set in screen pixels.

The Save button writes the object to the selected folder of My Places. The window stays open for the next object. The Clear button removes the points put so far.

Vertices are shown as circles and are dragged with the mouse. The circle in the middle of a segment adds a new vertex there. A click on the first point closes the shape, and a path becomes a polygon. A click on the last point finishes the path. A hint stands at the cursor. After finishing, clicks add no points, and the vertices can still be edited.

Right-click menu on the globe

A right click on the globe opens a menu.

Item	What it does
Delete vertex	Removes the vertex of the drawn object under the cursor
Finish drawing, Continue drawing	Stops and resumes adding points with clicks
Properties...	Opens the properties of the placemark under the cursor
Add placemark here	Opens the New placemark window with the point under the cursor
Fly here	Flies to the point with the same altitude and tilt
Copy coordinates	Latitude and longitude of the point to the clipboard

The shape of a 3D path, a 3D polygon and a tour is not edited this way. A placemark with a saved measurement loses the measurement text after its shape is edited.

Saved view

The Save view icon puts a placemark at the look-at point. The name is offered with a number - View 1, View 2. The placemark keeps the camera distance, heading and tilt. A flight to such a placemark brings the whole view back.

Place properties

Name

My path 1

Description

Color

Width

3.00

Height above ground

0.0 m

Ground

Space

Extend to ground

Time

None

2026-09-30 04:25

Place view

Latitude

58.035000°

Longitude

56.210000°

Range

3000 m

Heading

0.0°

Tilt

0.0°

Date/time

None

2026-09-30 04:25

Snapshot current view

Reset

OK

Отмена

The Properties... item opens the properties window. The window does not block the globe, the view can be turned and zoomed. Each change shows on the globe at once. While the window is open, the vertices of the placemark, path or polygon show on the globe as circles and can be dragged, a circle in the middle of a segment adds a vertex. OK writes the changes to the place together with the shape, Cancel brings the previous look and shape back.

Field	What it sets
Name	Name in the list and label on the globe
Description	Text of the placemark, it goes into KML too
Icon	Point icon from the QGIS set, in KML a standard icon of the same theme
Color, Width	Color of the point icon, color and width of the line and outline
Fill	Fill color of the polygon and of the wall
Height above ground	Lift of the object above the terrain in metres. The Ground - Space slider under the field sets it from the ground to 100 km
Extend to ground	A wall from the object to the ground, a post for a placemark
Time	Moment or interval of the placemark for the time slider
Place view	Look point, range, heading, tilt of the camera at the place and the view date

Extending works with a height above zero. A path becomes a wall, a polygon becomes a block.

Place view

Any place can have a view of its own. A view is the point the camera looks at, the range to it, the heading and the tilt. The look point may differ from the place. Fly to, a double click on the place and a tour stop follow the view.

Snapshot view in the place menu makes the current globe view the view of the place. The properties window has the same values in the Place view section. The Snapshot current view button takes the globe view, Reset brings back the view the place had when the window opened. Clearing the section check box removes the view, then the camera frames the whole place. The view goes into KML as a LookAt element and is read from it.

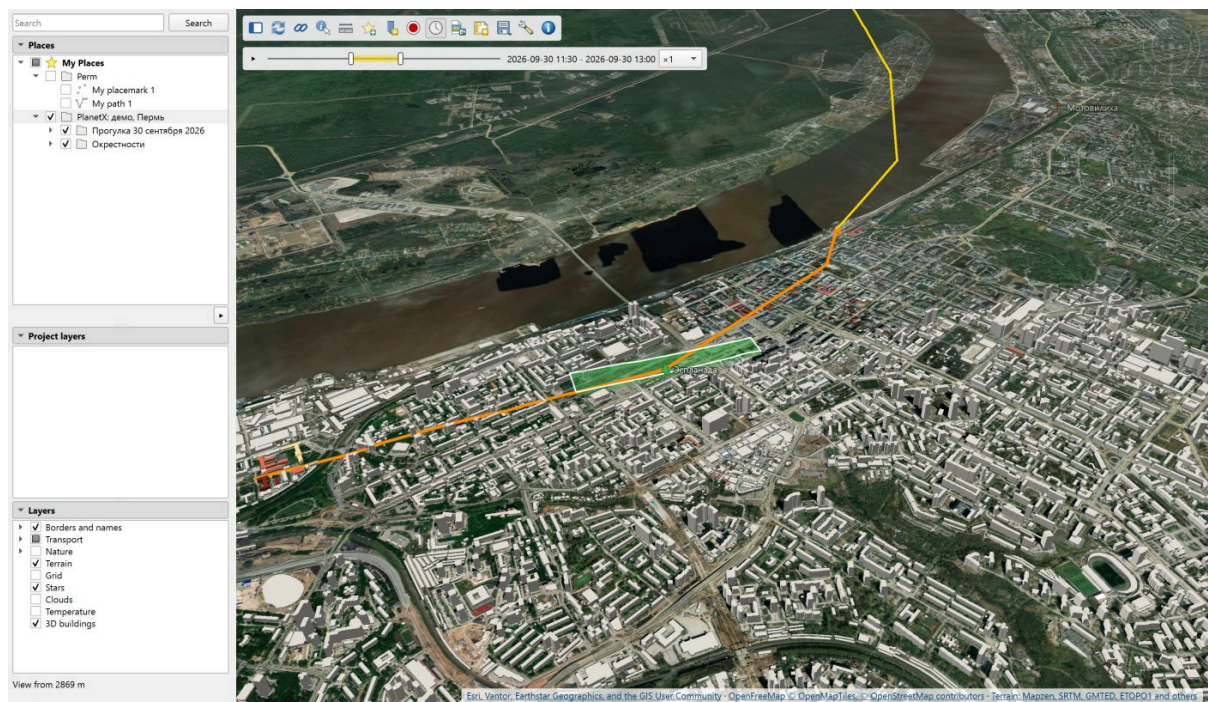
Copy and paste

Copy or Ctrl+C puts the selected places and folders into the clipboard as KML text. The text can be pasted into a text editor, edited and copied back. Paste or Ctrl+V puts the places from the clipboard into a folder. On a place it is the folder of that place, on empty space the root of My Places. Folders from the clipboard come in as folders. KML copied in Google Earth pastes the same way.

Placemark time and the time slider

The time slider is one time of the whole view. Coverages - NASA themes - take the moment, the right handle. Events - placemarks with time and earthquakes - take the range between the handles. When a theme is on and there are no events, the slider has one handle. The ◀ and ▶ buttons move the moment by a step of the theme series, ► shows the days of the series one after another and waits until each day has loaded. The legends of the layers - theme, foci, cutaway, beds, insolation, slope and temperature - stand as one panel in the bottom left corner of the view.

A placemark can have a time of its own. It is a moment or an interval, the Time field of the placemark properties. In KML the time is written as TimeStamp and TimeSpan. The view of a placemark has its own time, the Date/time field of the Place view section.



The time slider opens with the Time slider button on the icon bar. The button is available when visible placemarks have a time. The slider spans the placemark times from the earliest to the latest. A closed slider hides no placemarks, all of them show.

Part	What it does
Handles	Set the interval. Placemarks outside it are hidden, placemarks without a time always show
Bar between the handles	Dragging moves the whole interval, a click on the bar centres it on the click
⏮, ⏭	The interval of the same width moves to the start or the end of the slider
▶, II	Playback and pause, the interval moves along the slider
	Playback in a loop, after the end of the slider the interval starts from the beginning
Speed	Slider in 20 s crosses the whole slider in 20 seconds. $\times 1$ - time runs as on a clock, $\times 60$ - a minute per second, $\times 3600$ - an hour, $\times 86400$ - a day
Now	The moment goes to the computer clock, time runs at $\times 1$. The button is there when the Sun, satellites or the sky are on
■	Closes the slider

A flight to a placemark and a tour set the slider to the time of the placemark view, without it to the time of the placemark itself. A closed slider opens then. The slider is its own and does not depend on the QGIS Temporal Controller.

The Sun and Satellites rows and the sky view depend on the moment, not on data. With them the slider opens without timed placemarks too, on the computer clock, with one handle. The slider spans a day before the moment and a day after it. Playback at a clock speed does not stop at the edge of the span, the span moves after the moment. So at $\times 3600$ the line between day and night runs over the Earth, and the satellites pass their revolutions. A closed slider returns the Sun and the satellites to the computer clock.

KML and KMZ

Open KML or KMZ... puts a KML or KMZ file, including one from Google Earth, into My Places as a new folder named after the file. Folders, styles, icons, times and placemark views are kept, `gx:Tour` becomes a recorded tour. The camera flies to the contents of the file.

Save as KML... saves a folder or the whole My Places to KMZ or KML. The file extension chooses the format.

Image overlays

The Add submenu of a folder adds an image from a PNG, JPEG or GIF file in three ways:

- Ground overlay - the image lies on the terrain by four corners, in the middle of the view at half of its width. It lies under the borders and project layers.
- Photo - the image stands as a plane in 3D in front of the current camera. Fly to puts the eye at the camera point of the photo, and the photo covers the view.
- Screen overlay - the image stands in a corner of the view and does not move with the globe, for example a logo or a legend.

The overlay opens in a properties window. It changes the name, description, image and opacity, for a screen overlay also the corner of the view and the width. OK saves the changes, Cancel brings back the previous state.

A ground overlay is placed in one of two ways, as in KML:

- Box - north, south, east, west and rotation, fields of the properties window. While the window is open, handles stand on the globe. The centre moves the image, the handle above the north side rotates it,

the corners and the middles of the sides stretch it from the opposite corner or side, with Shift from the centre. A new image is placed as a box.

- Four corners - the Convert to four corners button. Each corner is dragged with the mouse on its own, the image can become an irregular quadrilateral. It does not convert back to a box.

The image is kept in one of two ways, the Storage field:

- In the places file - a copy inside My Places, shown without the source file.
- Link to a file or address - a file path or an http(s) address. The image is read when shown, an edit of the file shows on the globe.

The Refresh field sets the interval after which a linked image is read again while it is shown. A web address is requested past the cache, so the globe shows a fresh image or map that the server updates by itself. An interval shorter than 10 s is raised to 10 s. In KML this is `refreshMode onInterval` and `refreshInterval` of `Icon`.

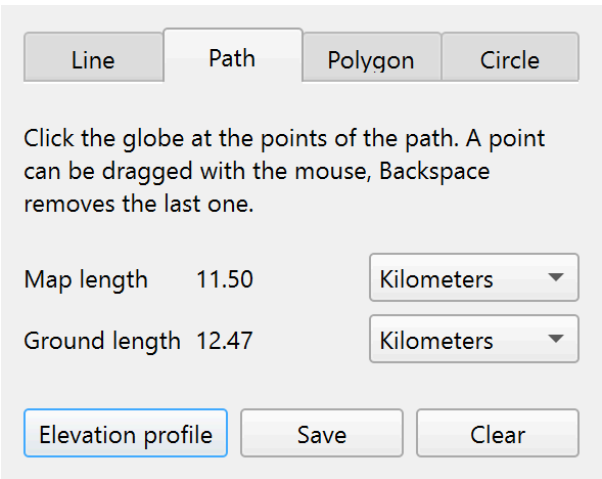
KML and KMZ with Google Earth overlays - GroundOverlay, ScreenOverlay and PhotoOverlay - open the same way. The image of a KMZ goes into the places file, the image of a plain KML stays a link to a file next to it or to a web address. Save as KML... to KMZ puts into the archive the images from the places file and the ones linked to files. To KML the images from the places file go into a folder next to it, links stay links.

Image to the QGIS project... in the menu of a ground overlay writes it to a GeoTIFF file in WGS84 by the corners of the image. The file is added to the project as a raster layer in the PlanetX - images group, with the opacity of the overlay. Ground overlays to the QGIS project... in the folder menu does the same for all ground overlays of the folder, into a chosen directory.

Clear My Places... removes all placemarks and folders after a confirmation. The removal cannot be undone, so the places needed are better saved to KML first.

Ruler

The Ruler icon opens a window with the tabs Line, Path, Polygon, Circle, 3D path and 3D polygon. Points are put with clicks on the globe, a rubber band follows the cursor.



Tab	Values
Line	Map length, ground length, heading. The third click starts a new line
Path	Map length, ground length
Polygon	Perimeter, area

Tab	Values
Circle	Radius, perimeter, area. The first click is the center, the second sets the radius
3D path	Length by straight segments in space
3D polygon	Perimeter, area in the plane of the polygon, tilt of the plane

Map length, perimeter and area are computed on the WGS84 ellipsoid. Ground length runs along the surface with its rises and falls. Heading is the azimuth of the start of the line clockwise from north. Each value has its own units - metres, kilometres, miles, nautical miles, and for the area square metres, hectares, square kilometres, square miles.

Heights for the ground length come from Mapzen Terrain Tiles. Missing height tiles along the line are loaded, even when terrain is off. While they load, the ground length is preceded by «≈».

A ruler point can be grabbed with the mouse and dragged, Backspace removes the last point. Each segment on the globe is labelled with its length.

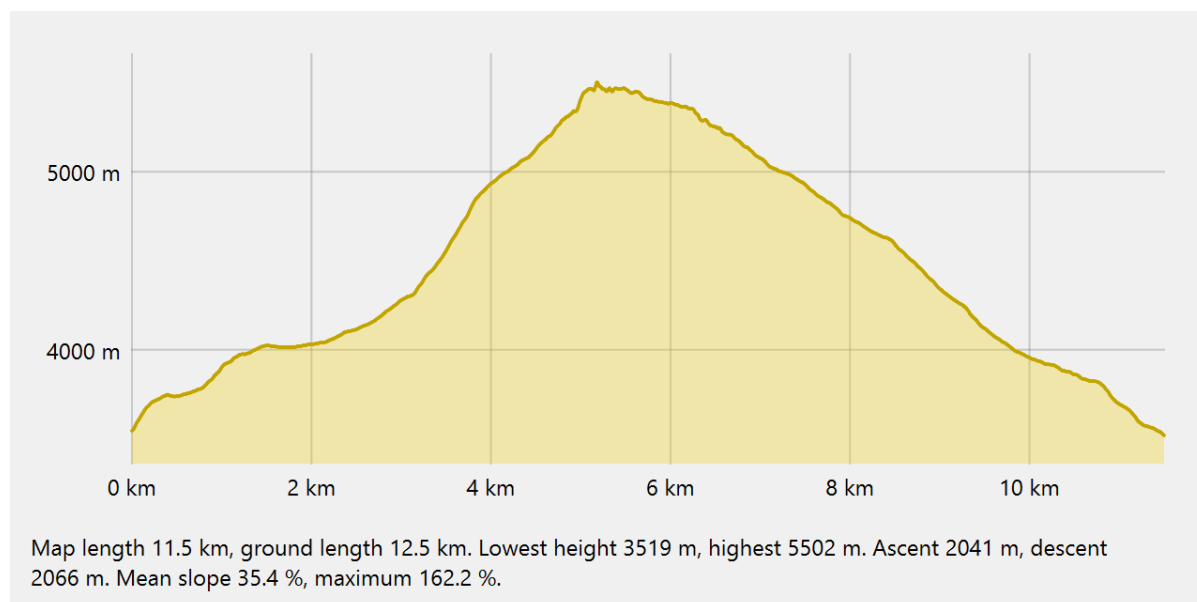
The Save button puts the shape into My Places together with the measurement. The name is offered with a number, for example Line 1. The measurement shows in the row tooltip.

The 3D path and 3D polygon tabs put a point on the roof or wall of a 3D building when the ray from the eye meets it before the terrain. They keep the height of the point. Segments run straight between the points and do not follow the terrain. The area of a 3D polygon is measured in its plane, so for a roof slope or a wall it is the real one. Tilt is the angle of the plane to the horizon, 0 is a flat roof, 90 a wall. A saved 3D shape goes to KML with altitudeMode absolute.

The Ruler and New placemark windows are open one at a time. Opening one closes the other.

Elevation profile

The Elevation profile button of the ruler opens a chart of height along the line. The Elevation profile item in the menu of a path in My Places does the same.



Under the chart are the map and ground length, the lowest and highest height, ascent and descent, the mean and maximum slope. The cursor over the chart shows the distance, height and slope at that place, and a mark with the height shows the point on the globe. The profile of the ruler follows its points.

Heights are taken as for the ground length. The slope is measured over a stretch not shorter than three pixels of height data, so roughness of the data does not pass for a cliff. On steep mountains the maximum slope may exceed 100 %, that is 45°.

Viewshed from a point

The Viewshed from here... item in the menu of a point placemark opens the Viewshed from a point window. The window sets the observer height above the terrain, the target height and the circle radius. The Build button loads the heights under the circle and colours it. Visible places are green, places hidden by the terrain are red. The Remove button removes the colouring.

Rays leave the point with an equal azimuth step. The step along a ray is one five-hundredth of the radius, 20 m for a radius of 10 km. A place is visible if the ray from the eye to it does not touch the terrain on the way. The computation takes the curvature of the body into account, and on the Earth the refraction with the coefficient 0.13. Buildings and forest are not taken into account. The window shows the visible share of the circle area. If the heights are not loaded within 30 s, the computation uses less detailed ones, and the window says so. The viewshed is computed on the Earth, Mars and the Moon.

Insolation

The Insolation... item in the menu of a point placemark opens the Insolation window. The window sets the first and the last day of the period and the circle radius. The Build button loads the heights and colours the circle by the mean number of hours of direct sunlight per day. Blue means little light, red means much light. The scale in the bottom left corner of the view runs from zero to the largest value in the circle. The Remove button removes the colouring.

The position of the Sun is computed every 10 minutes of the UTC day. A place is lit if the Sun is above the mountain horizon at its azimuth and falls on the front side of the slope. The horizon is taken in 32 directions at a distance of at least the radius and at least 5 km. A period longer than 15 days is computed over 15 evenly spaced days. Clouds, refraction, buildings and forest are not taken into account. The height grid has 800 nodes per side, the step is 25 m for a radius of 5 km. The computation runs in parts, and the globe responds to the mouse meanwhile. The line under the fields shows the share done. Insolation is computed only for the Earth.

Subsurface mode

The Subsurface mode icon on the icon bar opens a window of the same name. The mode shows drill holes, bed roofs, section walls and a block cut under the surface.

Data

The data are tables in the same layout as in Isoliner.

Table	Fields	What it sets
collar	hole_id, z, eoh, point geometry	Hole collar, collar elevation, end-of-hole depth along the hole
interval	hole_id, from, to, code	Intervals along the hole, positive downward, code is the bed or lithology
survey	hole_id, depth, azimuth, dip or zenith	Survey. Without it the hole is vertical
beds	code, ord, color, surface	Order of beds from top to bottom, colour "#rrggbb", roof raster file
sections	line geometry	Section lines

Table	Fields	What it sets
cut	polygon geometry	Block cut
images	name, image, top, bottom, line geometry	Image section: a PNG or JPG file relative to the folder of the GeoPackage file, elevations of the top and bottom of the image in metres
tunnels	name, depth, depth_end, diameter, color, line geometry	Tunnel: depth of the axis below the surface at the start and the end of the line, diameter in metres, colour "#rrggbb"

Fields are also found by other common names, for example bhid, elev, td, from_m, to_m, litho. A dip angle with negative values is counted from the horizontal, -90 is straight down. A bed roof is a raster of elevations in metres in any coordinate system, GDAL reads it.

Geological project layers show in 3D as ordinary globe layers. A layer is recognised by a table name from the list above or by its fields, for example a point layer with the fields hole_id and eoh is collars. A layer checked in Project layers goes under the surface, an unchecked box removes it. Such layers are not drawn as a flat image on the surface. A roof is a raster whose file is named in the surface field of the beds table. The interval, survey and beds tables are taken from the project by themselves, first from the same layer group.

The path to a section image is relative to the folder of the file of its layer. When the project has several layers of one role, the top one in the layer tree is taken.

The style comes from QGIS, as with ordinary layers:

- a roof whose raster is coloured with a pseudocolour ramp, a palette or RGB takes this colouring. A grey raster without settings takes the bed colour from the beds table.
- a hole without intervals and a tunnel take the colour of the feature symbol when the layer style is categorized, graduated or rule-based. With a single symbol the colour comes from the color field, without it the mode uses its own colour.
- the opacity of a QGIS layer is the opacity of its objects under the surface.
- the labels above the collars are the labels of the QGIS collar layer, from a field or an expression. With the layer labels off there are no labels.

After the style or data of a layer is edited, the globe rebuilds the model as with ordinary layers. The Create template... button of the Subsurface mode window writes a GeoPackage file with all tables of the mode and an example at the view point - a drill hole with two intervals, a section, a tunnel and a cut - and adds it to the project as the Subsurface template group. The template tables are filled with your own data.

Display

The model is built when at least one geological layer is checked. The globe loads the terrain heights under the model and puts it under the surface.

- A drill hole follows the survey by the minimum curvature method. Intervals take the colour of their bed, the layer label stands above the collar.
- A bed roof is a surface in the colour of the bed.
- A tunnel is a tube of its diameter along the line below the terrain. The depth changes from depth at the start of the line to depth_end at its end, this sets a ramp to a portal. Without depth_end the depth is constant, without diameter it is 3.7 m.
- A section is a set of walls along the line from the terrain to the model bottom, beds between the roofs have their own colours.
- An image section is a vertical wall along the line from the top elevation to the bottom elevation with the image stretched on it. The left edge of the image is the start of the line, its top is the top elevation.

An image larger than 4096 pixels on the longer side is scaled down.

- The model bottom is the end of the deepest hole, without holes it is 20 m below the deepest roof.

Depth is stretched by the terrain scale, like the surface. With the terrain off, elevations are counted from the surface.

The Surface opacity slider makes the terrain inside the model frame transparent, the drill holes and roofs show through it. Outside the frame the surface stays opaque. The Block cut checkbox removes the surface and the roofs inside the cut polygon. Walls with beds stand along the cut edge, a plane at the model bottom lies at its floor. The Camera under ground checkbox lets the camera go below the terrain down to the model bottom. Inside the model frame the camera then moves along its bottom. The window settings are kept in the project.

A polygon of My Places cuts the model with the Model cut item of its menu, a path puts a wall with beds with the Model section wall item. Choosing the item again removes the cut or the wall. Editing the shape of the place changes the model at once. So a cut and a section are drawn on the globe with the New placemark tool.

The Beds legend in the bottom left corner of the view shows the beds of the model from top to bottom in their colours and the tunnels.

In the Identify mode a click on a drill hole shows its number, the bed, the depth along the hole and below the collar, the elevation and the end of hole at the point under the cursor. A click on a tunnel shows its name, the depth and elevation of the axis and the diameter.

The Model section... item of the path menu in My Places opens the window of the same name. The horizontal axis is the distance along the path, the vertical axis is the elevation in metres. The section shows the terrain, the beds between the roofs, the drill holes in the colours of their intervals and the tunnels at their diameter from a band along the path. The Band of holes and tunnels field sets the band width. The cursor over the chart marks the point on the globe.

The subsurface demos put their data into the QGIS project as a group of ordinary layers and check them on the globe. The Permian deposits demo in the Demo icon opens a synthetic site near Berezniki, on the Verkhnekamsk deposit. It has 24 drill holes, 9 bed roofs of the Verkhnekamsk section, two sections, a cut of the north-east quarter and the image section 3-3. The data are made up for the example, they are not a survey.

The Vegas Loop tunnels demo opens the tunnels of The Boring Company under Las Vegas with their stations, about 8 km of lines. The routes and stations come from OpenStreetMap. OpenStreetMap has no depth, the tunnel axis is 12 m below the surface, about 40 feet after public information on the Las Vegas Convention Center tunnels, and ramps to the stations go from the surface down to this depth.

Tours

A tour flies over the checked places of a folder in the list order, including nested folders. The description of the stop place shows below the buttons of the tour bar.

- A place with a view of its own brings back its look point, range, heading and tilt.
- Along a path the camera travels the line.
- Other places are framed whole.

A tour starts with Play tour in the menu of My Places or a folder. A path has Tour along the path in its menu. The ► button under the places list starts a tour too. It tours the selected folder or path, and with a place selected it tours the folder holding the place. A tour goes through the places of the body now on the globe. When no places are checked, the status line says so.

The tour bar appears at the bottom of the view:

Button	What it does
Slider	Shows how much of the tour has passed and winds the tour. The camera moves to the chosen point at once, after release the tour goes on from it
⏮, ⏭	Previous and next stop
⏸, ►	Pause and continue
Pause	How many seconds the camera stays at a stop
🔄	Loops the tour. After the last stop the tour starts again from the first. The button state is kept between sessions
●	Records the tour as PNG frames
■	Ends the tour

Moving the camera with the mouse pauses the tour.

Recording a tour from the screen

The record button ● on the icon bar records the camera movement. The camera can be moved with the mouse, keys, flights and the navigation controls. A second click ends the recording and asks for a name. The tour goes into My Places with a film icon and is not drawn on the globe.

Play tour in the menu of such a tour plays the recording. The camera first flies to the start of the recording. The slider of the tour bar and recording to PNG frames work as for a tour over places. In KML a recorded tour is saved as `gx:Tour`, and Google Earth plays it.

Recording a tour for a video

The ● button asks for a folder and records the tour into it as PNG frames, 25 frames per second of the tour. The frame size equals the window size. The source credits stand in the corner of each frame. The files are named `frame_00000.png`, `frame_00001.png` and on, frames with the same numbers in the folder are replaced. Each frame waits for the tiles, so recording takes longer than the tour and the frames have no blurred spots. A frame that waits longer than 20 seconds is taken as it is. Recording always starts at the first stop, so a repeated recording gives the same frames.

A file `frames.json` lies next to the frames - frame rate, size, camera pose and data time of each frame, numbers of frames that did not fully load.

The bar shows the frame number. The cross on the bar stops the recording.

Making a video from the frames

PlanetX writes frames, not a video file. Any program that opens an image sequence assembles a video from them.

The free program ffmpeg (ffmpeg.org) makes an MP4 with one command in the folder with the frames:

```
ffmpeg -framerate 25 -i frame_%05d.png -vf "pad=ceil(iw/2)*2:ceil(ih/2)*2" -c:v libx264 -pix_fmt yuv420p -crf 18 tour.mp4
```

The rate 25 matches the recording rate, so the video runs at the pace of the tour. The `pad` filter adds a pixel when a frame side is odd, the H.264 codec needs even sides. A smaller `-crf` value gives better quality and a larger file.

The video editors Shotcut and DaVinci Resolve open numbered frames as one clip. Music, captions and several tours joined together can be added there.

Tracks

A track shows the path of moving objects over time. The source is a point layer of the project with a time field, and the time slider of the globe sets the time. The QGIS Temporal Controller does not act on the globe.

The Track... item in the menu of a point layer opens the settings window:

Field	What it sets
Time	A field with date and time, ISO 8601 text or seconds since 1970
Object	A field that tells the objects apart. One object - all points form one path
Color	Color of the travelled path
Camera follows	The camera keeps the first object in the middle of the view along the motion

The time of a track enters the extent of the time slider. Each object grows its travelled path up to the right handle of the slider, and a label with its name marks the current point. With the slider closed, whole tracks are shown. The Remove track button returns the layer to its usual look. Track settings are stored in the project.

During a tour recording the tracks grow along the tour over the range of the open slider, with the slider closed over the whole track time.

Project layers with time

A project layer with time switched on in its QGIS temporal properties is shown on the globe within the range of the time slider. The time of the layer enters the extent of the slider. The picture of the layer and its labels are redrawn after the handles stop. With the slider closed, the layer is shown in full.

Satellites

The Satellites row of the Layers section shows artificial satellites of the Earth as dots at their height. The groups stand under the row, any of them can be checked:

Group	What it holds
Space stations	the ISS, Tiangong and the docked spacecraft
Bright satellites	about a hundred satellites and stages visible without binoculars
Navigation	GPS, GLONASS, Galileo, BeiDou
Weather satellites	weather satellites
Earth observation	imaging satellites, Landsat and Sentinel among them
Science	science satellites and observatories

Group	What it holds
Geostationary	active satellites at about 36 000 km
Starlink, OneWeb	communication satellites, thousands of dots

CelesTrak gives the orbital elements of the groups. The position is calculated by the SGP4 model for the right handle of the time slider. With the slider closed, the satellites follow the computer clock. A satellite whose elements are older than 30 days at that moment is not shown.

CelesTrak updates the elements once in 2 hours and asks to download a group no more often. A downloaded group is kept in the QGIS profile, and a new request goes no earlier than 2 hours later. After a refusal of CelesTrak the group is not requested for 2 hours.

A click on a satellite dot with identification on opens the Identify window with the name, the NORAD number, the group, the height, the speed and the age of the elements. The satellite becomes the selected one.

The selected satellite shows its orbit loop at its height and its ground track. The loop shows one revolution ahead of the moment, the track shows the path of the point under the satellite half a revolution back and ahead. A right click on a satellite dot opens a menu with its name and two items:

- Orbit and ground track shows or removes the loop and the track.
- Camera follows puts the look point under the satellite, the view turned along its motion. The distance and the tilt of the view change with the wheel and the mouse as usual. Choosing the item again releases the camera.

Scenes

A scene saves the whole view and can be passed to other people. The Scene icon opens the menu Save Scene... and Open Scene.... A scene file has the extension `.planetx`.

A scene holds:

- the camera pose.
- the range of the open time slider.
- the project layers checked on the globe, as a link to their source.
- the base map, terrain, terrain exaggeration, groups of the Layers section and the label language.
- the rows of the Layers section, the grid, stars, clouds, temperature and 3D buildings.
- the selected folder of My Places with its tour.

A database password does not go into the file, a link to a QGIS connection setting stays. When the whole My Places folder or nothing is selected, no places go into the scene.

When a scene opens, a layer is looked up in the project by its source and added if missing. The other project layers are unchecked on the globe. The places of the scene come in as a new folder, and the camera flies to the scene pose. The status line lists the layers that were not found and did not open.

Demo

The Demo icon with an academic cap on the icon bar opens prepared scenes. Each scene puts its places into My Places as a new folder, the ► button under the list plays a tour over them. The folder can be deleted.

Section	Scene	What it holds
Earth	Perm	Places with icons and the moments of a walk, place views with a date, a route, an extruded polygon, a path along the Kama, a recorded flight over the centre, 3D buildings
Earth	Boca Chica, Starbase	The Starbase launch site and factory, the beach, nearby towns, the highway from Brownsville, a recorded flight around the launch site
Earth	Permian deposits	A synthetic site near Berezniki with drill holes, roofs, sections and a cut. A tour of five stops explains the subsurface mode, see Subsurface mode
Earth	Vegas Loop tunnels	The Boring Company tunnels under Las Vegas with their stations, see Subsurface mode
Earth	Aral Sea	MODIS images of 2000, 2014 and 2023 as a group of radio buttons, a photo of the ships at Moynaq and a basin map on the screen. All images are links to web addresses, see Image overlays
Earth	Quarry, own terrain	A synthetic 1 m survey of a quarry near Berezniki as the globe terrain: 10 benches of 15 m, a waste dump, a path for the elevation profile. The raster goes into the QGIS project, see Own terrain
Earth	Japan Trench	Earthquakes, the Earth cutaway with a sector from the equator to 38.5° N, the Slab2 slab under Japan, the epicentre of the 2011 Tohoku earthquake, Mount Fuji. The Section window along 38.5° N opens with the demo, the foci band is 300 km
Mars	Rover landing sites	Olympus Mons, Valles Marineris, the landing sites of Curiosity, Perseverance, Zhurong, Spirit and Opportunity
Mars	Jezero crater	The Perseverance landing site, the surroundings of the crater, an elevation profile across the crater, a point on the rim for the viewshed, a flight around the crater, the Slope layer and terrain with scale 3
Moon	Apollo and Lunokhod sites	The landing sites of six Apollo missions, Lunokhod 1 and Lunokhod 2
Sky	Constellations and bright sky objects	Orion, the Pleiades, the Andromeda Galaxy (M31), Cassiopeia, Ursa Major, Lyra with Vega, the Southern Cross

The Perm walk time is fictional. Landing coordinates are rounded to hundredths of a degree.

View snapshot and view to layout

View snapshot

The View snapshot icon opens a window with the snapshot width and height in pixels. A snapshot can be larger than the window, the globe loads detailed tiles for it. The Window proportions check box keeps the height in step with the width.

The Save... button asks for a PNG or JPEG file and starts the snapshot. The camera stands still meanwhile. The Take now button does not wait for the remaining tiles. The source credits stand in the bottom right corner of the snapshot.

View to layout

The View to layout icon puts the view as a fixed picture into a QGIS layout, an existing one or a new one.

Field	What it sets
Layout	An existing layout or New layout
Width, Height	Size of the picture on the page in millimetres

Field	What it sets
Resolution	Output resolution of the layout, it sets the snapshot size in pixels

The picture is kept inside the project. The layout window opens after the insertion.

Other bodies

The Body icon on the icon bar opens the menu of bodies. The Earth is at the top, the Moon and Mars are below a separator, then come submenus:

Submenu	Bodies
Planets	Mercury, Venus, Jupiter
Moons of Jupiter	Io, Europa, Ganymede, Callisto
Moons of Saturn	Mimas, Enceladus, Tethys, Dione, Rhea, Titan, Iapetus
Asteroids	Ceres, Vesta

The menu holds only bodies imaged by spacecraft in full. On Pluto, Charon and Triton 24 to 33 % of the globe is not imaged, they are not in the menu.

The chosen body replaces the Earth. The size of the globe, the imagery and the atmosphere change, and the camera flies to the start point of the body. On Mars it is Olympus Mons, on the Moon it is the Sea of Tranquility, the Apollo 11 landing site, on Mercury the Caloris basin, on Venus Maxwell Montes. On other bodies it is a point on the equator and the prime meridian.

Imagery of the bodies other than Mars and the Moon comes from global mosaics of USGS Astrogeology and NASA, made from MESSENGER, Magellan, Galileo, Voyager, Cassini and Dawn data. A tile pixel at the equator ranges from 100 m on Enceladus and Vesta to 120 km on Jupiter. Some imagery does not look as the eye would see it:

- Venus is shown by the Magellan radar, the colour shows the height
- Jupiter is a Cassini cloud map of December 2000, its clouds do not move
- Titan is imaged in near infrared light through the haze
- Mercury, Io and Ganymede are in colour, other bodies are grey

The globe of each body is a sphere, Jupiter is an oblate ellipsoid. Vesta, Ceres and small moons are not spherical in reality, their imagery is slightly distorted on the sphere.

Mars imagery is the Viking MDIM2.1 colour mosaic, about 650 m per pixel. Moon imagery is the LOLA albedo map with hill shading, about 670 m per pixel. Mars has a thin, dusty atmosphere, the Moon has none.

Mars terrain is built from the MOLA MEGDR heights, Moon terrain from the LOLA GDR heights. Heights are rounded to 10 m on Mars and to 20 m on the Moon. A height pixel at the equator is 2.6 km on Mars and 1.3 km on the Moon. The terrain is switched on with the same Terrain row of the Layers section, the vertical scale is set in the view properties. Mars heights refer to the areoid, Moon heights refer to a sphere of radius 1737.4 km. Depressions below zero are kept, the floor of the Hellas plain lies 7-8 km below zero.

On other bodies the navigation, grid, stars, ruler, My Places, tours, view snapshot and scenes work, on Mars and the Moon terrain works too. Functions with Earth-only data are off:

- vector base, clouds, temperature, 3D buildings and sunlight
- project layers and their labels, tracks

- synchronization with the QGIS map and feature identification
- place search by name. Coordinates in the search field work

Each place remembers its body. My Places lists the places of all bodies, the globe shows only the places of the current one. A flight to a place on another body switches the body first. A tour goes through the stops of the current body. A scene keeps its body and opens on it.

Assistant

The assistant answers topics and requests from the Search field. Its buttons stand in the Assistant group of the View properties window. Conversation... opens the Assistant window, Settings... the Assistant settings window. The conversation window with an AI model is Assistant. The model controls the globe on a request in words, for example "show a section across the Japan Trench" or "where are the deepest earthquakes near Japan this month". An API key of your own for the chosen service is needed.

The conversation goes in the Assistant window. The line at the top names the service and the model, the Settings button opens the Assistant settings window. There the service, address and model are chosen, the key is entered and the check boxes are set. By default OpenRouter with a free model is chosen.

- **Service.** Each service has its own key, address and model.
 - Claude (Anthropic) - Claude models, by default `claude-sonnet-5-5`.
 - Grok (xAI) - Grok models, by default `grok-4.7`.
 - DeepSeek - DeepSeek models, by default `deepseek-flash`.
 - OpenRouter (free models available) - models of many developers with one key. Models with `:free` in the name are free, the number of requests per day is limited. By default `openrouter/free`, it picks a free model that is available itself. If the chosen `:free` model is busy, the request goes to `openrouter/free`.
 - Own service, for example Ollama - any service of the OpenAI Chat format. A service on this computer needs no key. By default the Ollama address `http://localhost:11434/v1` and the model `qwen3`. The model must be able to call tools, otherwise the assistant only answers with text.
- **Key.** The key is entered in the Key field of the Assistant settings window and saved with the Save key button. The Keep the key without the master password box is checked by default. With it the key is saved as plain text in the QGIS profile settings, the master password is not asked. Anyone with access to the profile folder can read such a key. Without the box the key goes to the QGIS password manager, and QGIS may ask for the master password.
- **What the assistant does.** Flight to a point and place search, body change, rows of the Layers section and the time slider, the Earth cutaway and a section down along points, point information and an earthquake summary in a box.
- **Placemarks in KML.** The assistant proposes placemarks, paths, polygons, folders and tours as a KML document. Below the conversation a proposal appears with the Write to My Places and Cancel buttons. Nothing is written without the button. Placemarks selected in My Places can be read by the assistant as KML and returned corrected.
- **What goes to the model server.** The request text, the look-at point, the body and the rows switched on in the Layers section. Names of project layers and My Places do not go, except placemarks selected by the user.

The assistant's actions show in the conversation as Action lines. It makes no more than six rounds of actions in a row, then it stops.

Places from a description

A topic in the Search field, for example "voyages of Columbus" or "battles of the Hundred Years' War", becomes placemarks if no place with that name is found. Ctrl+Enter in the field and the Make places button of the Assistant window make placemarks at once, without a place search. The model answers with one KML document, it is written to My Places as a new folder at once, the camera flies to the places.

The model receives the rules of the document. They list everything the globe shows in space and in time:

- Places and events come as points, routes, fronts and borders - as lines along the real path, territories - as polygons.
- Time stands on every placemark with at least a known date or year. An event gets a moment, a campaign, a siege or a reign - an interval.
- A route is split into legs. Each leg is a line with its own time interval, on the time slider it grows from start to end.
- A change of a territory or a front line comes as several shapes with intervals in a row, on the time slider they replace one another.
- The document holds 100 placemarks at most.

The answer comes as a stream. While it comes, the line below the field shows the count of placemarks received and the Stop link. Stopping keeps the placemarks received whole. The plugin does the same when the answer breaks off or hits the length limit. The answer time grows with the size of the document.

The finished answer below the field names the folder, the number of placemarks and the number of placemarks with time. The time slider opens on the interval from the first to the last date. The Cancel link removes the made folder.

A topic like "World War I" searched with Enter finds places with these words in the name. Then the link Make places on the topic stands below the field, it does the same as the button.

The coordinates and dates come from the model, they need checking.

Requests from the Search field

A request can also be typed in the Search field of the left panel. The assistant receives a request with a question mark, a request of six words or more and a request that starts with a request or question word, for example "show", "where", "how many". A topic for which Nominatim finds no places, for example "Voyages of Columbus", turns into placemarks at once, as with the button next to the field. A question for which no places are found goes to the conversation. Outside the Earth any request except coordinates goes to the assistant. The answer appears below the field, the Conversation link opens the Assistant window with the history. Placemarks that the assistant proposes in the conversation are shown below the field with the Write to My Places link. The Answer requests from the Search field box of the Assistant settings window switches this off, then the field finds only places and coordinates. Without a saved API key the field also finds only places.

Starry sky

The Sky item of the Body menu shows the sky from the centre of the celestial sphere. It holds the Milky Way, 5080 stars down to magnitude 6, lines and names of 88 constellations and names of bright stars. The Sun, the Moon and the planets from Mercury to Neptune are there too. The north celestial pole is up and east is on the left, as on a star chart.

Action	What it does
Drag with the left button	The sky follows the cursor
Wheel, PageUp, PageDown, plus, minus	Changes the field of view from 2° to 110°
Double click with the left button	The point moves to the centre, the field of view halves
Double click with the right button	The field of view doubles
Arrows	The view moves by a tenth of the window height

The status bar shows the field of view, the right ascension and the declination of the point under the cursor. Equatorial coordinates refer to the epoch J2000.0. The Sun, the Moon and the planets are placed at the end of the open time slider range, otherwise at the computer clock time. Planet positions differ from the JPL Horizons ephemerides by at most 0.2° in the years 1800-2050, the Moon position by at most 0.3°.

The Constellations item of the same menu hides and shows the constellation lines and names. Stars, their names, the Sun, the Moon and the planets stay.

Places and tours work in the sky. Save view puts a place with the view direction and the field of view. New place puts a point with a click on the sky. Record tour records the motion over the sky. Sky places are kept in My Places with the others and are shown in the sky as yellow circles with a label. A tour and the ► button in the sky go through the sky places, between distant points the field of view widens on the way. A flight to a sky place from the list opens the sky.

The ruler, synchronization and feature identification are off in the sky. A view snapshot saves the sky with its labels. A scene keeps the view direction and the field of view. To return to a body, choose it in the same menu.

Limitations

- Project layers lie on the terrain as a picture, a layer cannot be raised or extruded by a field. Your own places have lift and extrusion.
- A layer with time from an expression does not enter the extent of the time slider, its range is set with the handles manually.
- Above latitude 85° the poles are covered with the ocean color, the Web Mercator tile grid ends there.
- Recording a tour takes longer than the tour itself.
- Buildings are blocks with flat roofs, roof shapes and facades are not shown.
- Mars and Moon heights are coarser than Earth heights, a height pixel at the equator is 2.6 km and 1.3 km. Craters smaller than 5-10 km do not show in the heights.
- KML does not store the body of a place, places from a KML file go to the current body.
- The sky is shown from the centre of the Earth, without the horizon of an observing site. Coordinates are not reduced to the date, by 2026 precession shifts them by about 0.36°.

Data sources

Imagery - Esri World Imagery, Esri sets the terms. Map data © OpenStreetMap contributors. Vector tiles and 3D buildings - OpenFreeMap. Terrain - Mapzen Terrain Tiles, SRTM, GMTED, ETOPO1 and other data. Clouds - NASA GIBS, VIIRS imagery. Temperature - NASA GIBS, MODIS and GHRSSST MUR. Stars - the Yale Bright Star Catalogue. Milky Way - NASA/Goddard Space Flight Center Scientific Visualization Studio, Gaia DR2: ESA/Gaia/DPAC. Place search - Nominatim. Mars - NASA, USGS, Viking MDIM2.1, the Moon - USGS, LRO LOLA, tiles of both bodies - OpenPlanetaryMap. Mars heights - NASA MGS MOLA MEGDR, Moon heights -

NASA LRO LOLA GDR. Imagery of other bodies - global mosaics of USGS Astrogeology and NASA Photojournal from data of NASA, JPL, DLR and mission institutes, the credit of each body is in the corner of the view. Constellation lines, names and star names - d3-celestial, © Olaf Frohn. Planet positions are computed from JPL orbital elements, the Moon position from the formulae of the Astronomical Almanac.

The terms of all sources are described in [SOURCES.md](#). Tiles go through the QGIS network settings and cache, the plugin keeps no cache of its own.

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